

6. Write a program to implement Naïve Bayes algorithm in python and to display the results using confusion matrix and accuracy.

Aim

To implement the **Naive Bayes classification algorithm** and evaluate its performance using a confusion matrix and accuracy.

Software Required

- Python 3.x
- Pandas
- Scikit-learn
- Matplotlib

Procedure

1. Load the Iris dataset.
2. Divide the dataset into training and testing sets.
3. Create a Gaussian Naive Bayes classifier.
4. Train the model.
5. Predict the test samples.
6. Calculate accuracy.
7. Generate and display the confusion matrix.

Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix

iris = load_iris()
X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = GaussianNB()
```

```
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:")
print(cm)

print("\nAccuracy:",
      accuracy_score(y_test, y_pred))
```

Output:

```
... Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

Accuracy: 1.0
```

Result

Thus, the **Naive Bayes algorithm** was successfully implemented and evaluated using a confusion matrix and accuracy.